Cesare Barbera

PhD Candidate in Artificial Intelligence

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cesbar99.github.io
Cesbar99
Cesare Barbera
Google Scholar

Research Interests

My research aims to understand how probabilistic information produced by machine-learning models can be made reliable and translated into better decisions. The goal is to design new methodology with rigorous guarantees for settings in which information is incomplete, costly or acquired sequentially.

Academic Experience and Education

2023–Present **PhD in Artificial Intelligence**, *University of Pisa*
Research on uncertainty quantification, reliable machine learning and human–AI decision–making

2020–2022 **MSc in Data Science**, *University of Trento*, Grade: 110/110 cum laude
with thesis: Exploiting Impact versus posterior in machine learning: an exploratory study. Advisors: Professor Andrea Passerini, Professor Katya Tentori

2017–2020 **BSc in Economics and Management**, *University of Trento*, Grade: 109/110
with thesis: COVID-19: A Critical and Empirical Analysis of Fatality in Relation to Age and Gender*. Advisor: Professor Giuseppe Espa

Industrial Experience

02/2022–06/2022 **Data Scientist Intern**, *Reale Mutua Assicurazioni*
Data cleaning and exploratory analysis of insurance and climate-related risk data, to identify patterns in weather-related vehicle accidents to support improved risk assessment and policy design.

Grants and Research Projects

2023–Present **National PhD Scholarship in Artificial Intelligence**

2023–Present **TANGO – Horizon Europe**, *Grant Agreement No. 101120763*
Research on hybrid decision-making paradigms.

2023–Present **FAIR – Spoke 2: Human-centric AI**
Research on interpretable and trustworthy machine learning under the supervision of Professor Andrea Passerini.

Papers (more on Google Scholar)

[1] **C. Barbera**, L. Stradiotti, L. Cascioli, L. Perini, N. Tax, D. Haimovich, J. Davis, A. Passerini, A. Pugnana. *A CaLLM Judge is a Better Judge: Mitigating Bias in LLM-as-a-Judge. Under review at ICLR 2027.*

[2] C. Giacchetta, A. Bogani, **C. Barbera**, G. De Toni, M. Caprio, A. Pugnana, A. Passerini. *Should I Stay or Should I Show? Learning to Selectively Disclose Information. Under review at ICLR 2027.*

[3] **C. Barbera**, L. Perini, G. De Toni, A. Passerini, A. Pugnana. *Divide et Calibra: Multiclass Local Calibration via Vector Quantization.* NeurIPS 2026.

[4] **C. Barbera**, L. Perini, G. De Toni, A. Passerini, A. Pugnana. *Multiclass Local Calibration with the Jensen–Shannon Distance.* AISTATS 2026.

[5] I. B. Schlicht, B. Sayin, Z. Zhao, F. M. Labonté, **C. Barbera**, M. Viviani, P. Rosso, L. Flek. *Zoom In Disparities in Healthcare LLM Q&A.* NLDB 2026.

Academic and Research Activities

Reviewer for ICLR 2027; evaluator for the ELLIS PhD Program pre-screening, 2026.

Supervisor of 1 Master Thesis.

Projects

Implementations of my research papers is open – github.com/Cesbar99.

Programming and Software Skills

Python (PyTorch, TensorFlow, scikit-learn, NumPy, Pandas), R, SQL, LaTeX, Git.